

Job Market Analysis

Transforming raw job data into meaningful insights for better career decisions



Project Overview

This project analyzes job market data to understand hiring trends, required skills, salary distribution, and job availability across different locations.

The goal: convert raw and unstructured job data into meaningful insights that help students and job seekers make better career decisions.



Job Market

Job Title	Company	Salary	Recationl	Skills	Experience	Job Tyed	Rating
Job Fratlon						110	\$981%
Job Finap		996	2.83	2.93	31.9%	203	11.1%
		288	310		36.9%	313	\$97
Jibb Tingpe	Sado%		13.8		\$181	12	2,01.81%
Sob'lHeape	Sab%	\$57	3.01	\$61	23.9%	49	2,373.9%
Johi Type	Rabr%	\$55	2.86	1986	13%	21	3,0,58%
Rob Plloas		\$90	1318		121%	9	1,53.51%
Job litfallee		256	3378	75	16%	30	\$60.9%
Rx16 Kiaahs		259	3419	33		30	189.9%
		390	33.3	32	18.9%	20	70.9%

Dataset Summary



Job Information

Job Title, Company, Location, Salary, Experience Required, Skills, Job Type, Rating



Data Issues

Missing values, inconsistent formats, duplicate entries, multiple skills in single column

Data Cleaning & Structuring

The raw dataset was first cleaned and structured using Microsoft Excel.

01

Handled Missing Values

Filled missing salary using average, filled missing job titles and skills

03

Removed Duplicates

Identified and removed repeated job entries

05

Structured Skills

Prepared data for further analysis

02

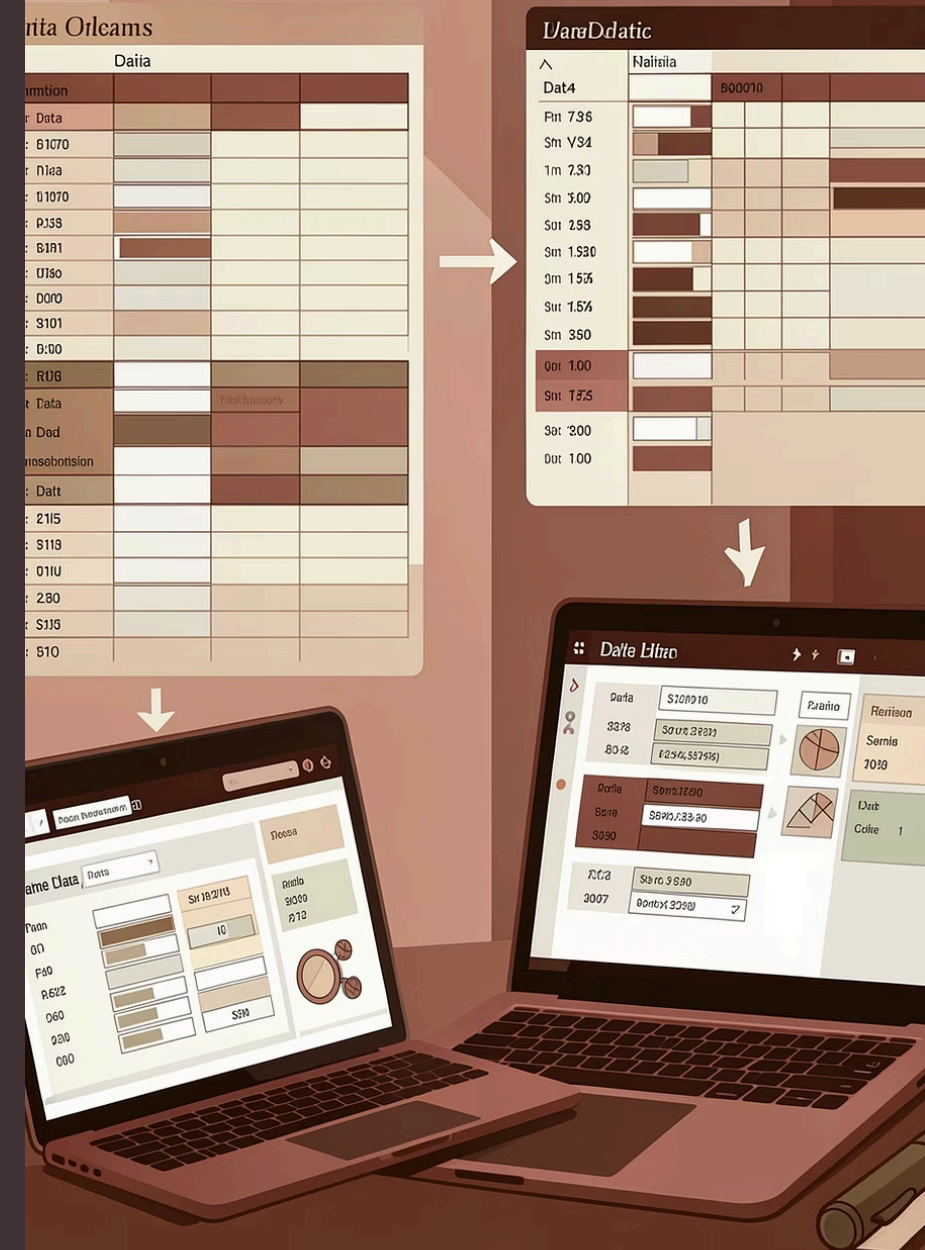
Standardized Salary

Converted formats like "₹6 LPA", "6 lpa" into numeric values (600000)

04

Cleaned Text Data

Converted job titles to consistent format (e.g., "Data Analyst")

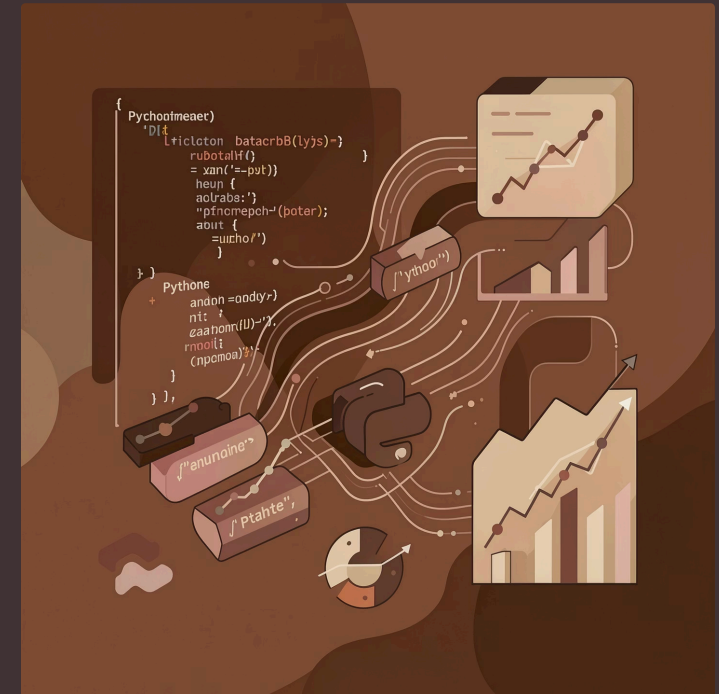


Data Analysis using Python

After cleaning, the dataset was analyzed using Python in Google Colab with the Pandas library.

Analysis Performed

- Job Role Demand - counted jobs per role
- Salary Analysis - calculated average and maximum salary
- Top Skills Analysis - split and analyzed skills
- Location Analysis - counted jobs by city
- Job Type Analysis - compared Full-time, Internship, Contract



Dashboard in Power BI

KPI Cards

Total Jobs, Average Salary,
Maximum Salary

Visualizations

Job Role Demand, Average
Salary by Role, Top Skills,
Jobs by Location, Job Type
Distribution

Filters (Slicers)

Location, Job Type, Job Title



Key Insights



Data Analyst is the most in-demand job role



Python, SQL, and Excel are the most required skills



Bangalore and Mumbai have the highest number of job opportunities



Machine Learning roles offer higher salaries



Full-time jobs are more common than internships

Real-World Problem



In today's job market, students and job seekers struggle to understand:

- Which roles are actually in demand
- What skills are required by companies
- How salaries differ across roles and locations

Most job data is unstructured and scattered, making it difficult to analyze. People often rely on assumptions rather than real data.

Solution & Impact



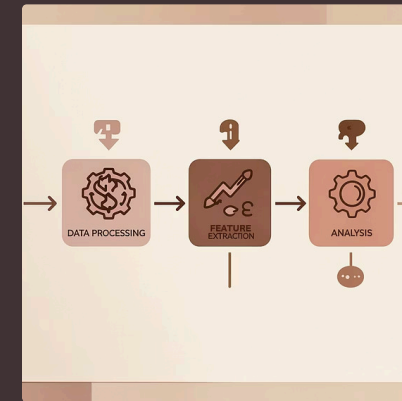
For Students / Job Seekers

Identify which skills to learn, gain clarity on career paths, make data-driven decisions



For Companies / Recruiters

Understand hiring trends, identify required skills, support better hiring strategies



For Data Analysis Learning

Demonstrates complete workflow: Data Cleaning (Excel), Analysis (Python), Visualization (Power BI)

Complete End-to-End Solution

This project demonstrates how raw job data can be transformed into useful insights through proper cleaning, analysis, and visualization.



Excel

Data Cleaning



Python

Data Analysis



Power BI

Data Visualization

